N v T STEP DIAGRAM based ANALYSIS in M/M/1 QUEUE SIMULATION

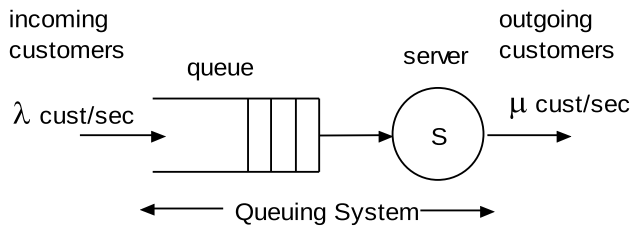

Fig.1 An M/M/1 queueing system

1. Step diagram for N vs time in an M/M/1 Queuing System

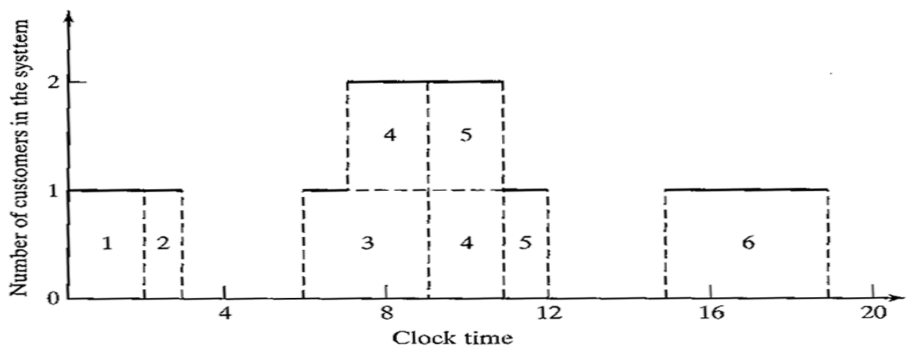

Fig.2 Number of Customers in the system vs time [1].

2. Simulation Table Showing Clock Times for an M/M/1 Queue

A	B	1	C	2	D	E	3
Customer#	Arrival Time (Clock)	IAT (Activity)	Time Service Begins (Clk)	Time Spent in Queue (Act)	Service Time (Activity)	Service Time Ends (Clk)	Time in System (Act)
1	0	0	0	0	2	2	2
2	2	2	2	0	1	3	1
3	6	4	6	0	3	9	3
4	7	1	9	2	2	11	4
5	9	2	11	2	1	12	3
6	15	6	15	0	4	19	4
SUMS		15		4	13		17
Mean IAT		3.00		Mean T{serv}	2.1667		
Lambda		**0.3333**		**Mu**	**0.4615**		

3. Calculations using the Simulation Table

QUESTIONS

Q1. Find the Mean Service Time for 6 customers given in the table.
Q2. Find the Mean Number in the Queue
Q3. Find the Mean Number in the System
Q4. Find λ, Mean Arrival Rate (customers/sec)
Q5. Find μ, Mean Service Rate (Customers/sec)

ANSWERS

Q1. T_s = 13/6 = **2.1667** time units.

Q2. ***Mean Number in Queue, Q*** = { 0 x (15) + 1 (4) } / 19 = 4/19 = **0.21** customers.
Evaluated as: In the system there were 0 customers in the queue for 15 time units, 1 customer was in the queue for 4 time units; giving a total of 4 customer x time units. Dividing this with 19 time units, we can find the mean number in queue.

Q3. Mean Number in the System, *N* = { 0x(6) + 1 x (9) + 2 x (4) } / 19 = 17/19 = **0.8947** customers.
Evaluated as: In the system there were 0 customers for 6 time units, 1 customer for 9 time units and 2 customers for 4 time units. The total number of *custom*er **x** *time-units* is found (17) and if divided by the total time elapsed (19 *time-units*) = 0.8947 customers.

Q4. Mean Arrival Rate, λ = No. of customers arrived /time elapsed = 6 / 19 = **0.316** cust/sec. (time-units)
NOTE: that if we use IAT to calculate λ using λ = 1/{IAT} gives slightly higher value since only 6 customers are considered and the time elapsed from 15 time units to 19 time units is disregarded. This gives a biased value for λ=1/3 = 0.333 cust/sec. Also IAT has to be calculated using 5 customers since the 1st customer arrives at t=0.

Q5. Mean Service Rate = 1/T_s = 1/2.1667 = **0.4615** cust/sec. (time-units)

We can also calculate the **Mean Time in the System, *T*** = 17/6 = 2.833 time units. ($RT=N/\lambda$ = 0.8947/0.316 = 2.831 time units.)

Further, **Mean Waiting Time in the Queue, W_q** = 4 time units/6 customers = 0.6667 and $RT = W_q + T_s$ = 0.6667 + 2.1667 = 2.833

Ref. 1.: J. Banks, et al, *Discrete-Event System Simulation*, 5th Ed., Prentice-Hall, 2010.